完整流体方程：

离子+电子：$\rho = n_i q_i + n_e q_e$ ← cold ion assumption：$n_i = n_0$.

$\rho = q_i n_i - q_e n_e = (n_0 - n_e)e$

离子很重，不以所包含频率运动，形成一固定均匀正电荷本底.

$\vec{j} = n_i q_i \vec{v}_i + n_e q_e \vec{v}_e$

Maxwel eqs：$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} = \frac{(n_i - n_e)e}{\varepsilon_0} = -\frac{n_{e1} e}{\varepsilon_0}$

$\nabla \times \vec{E} = -\frac{\partial}{\partial t}\vec{B}$ ✗ 纵波.

$\nabla \cdot \vec{B} = 0$. ✗ $\vec{B}_1 = 0$

$\nabla \times \vec{B} = \mu_0(n_i q_i \vec{v}_i + n_e q_e \vec{v}_e) + \frac{1}{c^2}\frac{\partial}{\partial t}\vec{E}$. ✗ $\vec{B}_1 = 0$

流体动力学方程：无热运动，$KT = 0$.

$m_j n_j [\frac{\partial \vec{v}_j}{\partial t} + (\vec{v}_j \cdot \nabla)\vec{v}_j] = q_j n_j (\vec{E} + \vec{v}_j \times \vec{B}) - \nabla \cdot \vec{P}_j + f_{\text{碰撞}}$ $j = i, e$.

高次项=0　　压力 $P = nKT$

连续性方程：

$\frac{\partial n_j}{\partial t} + \nabla \cdot (n_j \vec{v}_j) = 0$. $j = i, e$.

仅剩 $n_0 \nabla \cdot \vec{v}_{e1}$

状态方程：绝热：$\gamma = (2+N)/N$. 一维 $\gamma = 3$.

$P = C(m_j n_j)^{\gamma_j}$, $j = i, e$.

$\gamma_i = \frac{C_p}{C_v}$　　无热运动

$$\nabla \times \vec{E}_1 = -\frac{\partial}{\partial t}\vec{B}_1 + \quad 4\pi$$